

CHAPTER 1

INTRODUCTION

Background of the Study

Plastic pollution represents a worldwide challenge, with anticipated increases in plastic waste surpassing efforts to prevent (Borrelle et al., 2020).

Beyond the visible problem of plastic waste on land and its effects on society and the economy, it is now estimated that there are more than 170 trillion plastic particles floating in the oceans globally (Eriksen et al., 2023).

During the last ten years, there has been an increasing recognition and worry regarding land-origin plastic litter making its way into coastal and marine ecosystems, particularly in the Asia Pacific region. Many modeling studies have predicted this region to be an unequal contributor of improperly managed land-sourced waste transitioning from land to sea (Jambeck et al., 2015; Lebreton et al., 2017; Meijer et al., 2021).

The Philippines are projected to generate 7,415,202 tons of mismanaged waste in 2025 and had been ranked the 3rd biggest contributor to plastic pollution globally by nations that consist of coastal population, estimations of waste generated, the share of plastic in solid waste, and mismanaged solid waste (Jambeck et al., 2015).

In their 2019 report, state auditors identified waste management issues in the disposal of wastes in five hospitals in Pangasinan province namely Alaminos City, San Carlos City, Tayug, Mangatarem, and Lingayen. The report highlighted the need for appropriate control of infectious waste where materials such as vials, needles, ampules, used gloves and intravenous lines were not always disposed of through a designated transport facility (Rappler, 2020).



The Plustic Vendo Machine is one of the best ways to minimize plastic waste, and encourage recycling, and will be rewarded using a Reverse Vending Machine (RVM). By creating this machine using Arduino and becoming a convenient way for people to recycle plastic bottles. This project seeks to encourage recycling the students of the Golden West Colleges. Instead of dumping in land fill, road sides, oceans and rivers, the Reverse Vending Machine (RVM) gathers and stores plastic for recycling. The consumers are rewarded with receipts that earn them points, which makes recycling a rewarding experience. Individuals are encouraged through the reward system to change their mind set in handling waste, to become proactive in minimizing plastic waste.

Conceptual Framework

The developers used the Input-Process-Output (IPO) to present the paradigm of the study that illustrates the connection between the input process and output.

The Conceptual Framework is reflected in the paradigm of this study, as shown in Figure1. The following are the identified inputs needed in creating a Plustic Vendo Machine. The inputs required to operate the Plastic Reverse Vending Machine, which include empty PET bottles inserted by users and the availability of receipt with points for rewards. The process involving the detection and validation of PET bottles, counting mechanisms, activation of the points system, and storage management of collected bottles. And the features of the device.

The process includes the RAD as the software life cycle model in developing the Plustic Vendo Machine. The software methodology phases that will be involved are defining the project requirements, prototyping, rapid construction and feedback gathering, and finalizing the product and implementation.



After undergoing such processes, the output of the study is the Plustic Vendo Machine. Figure 1 on the next page shows the research paradigm of the study.

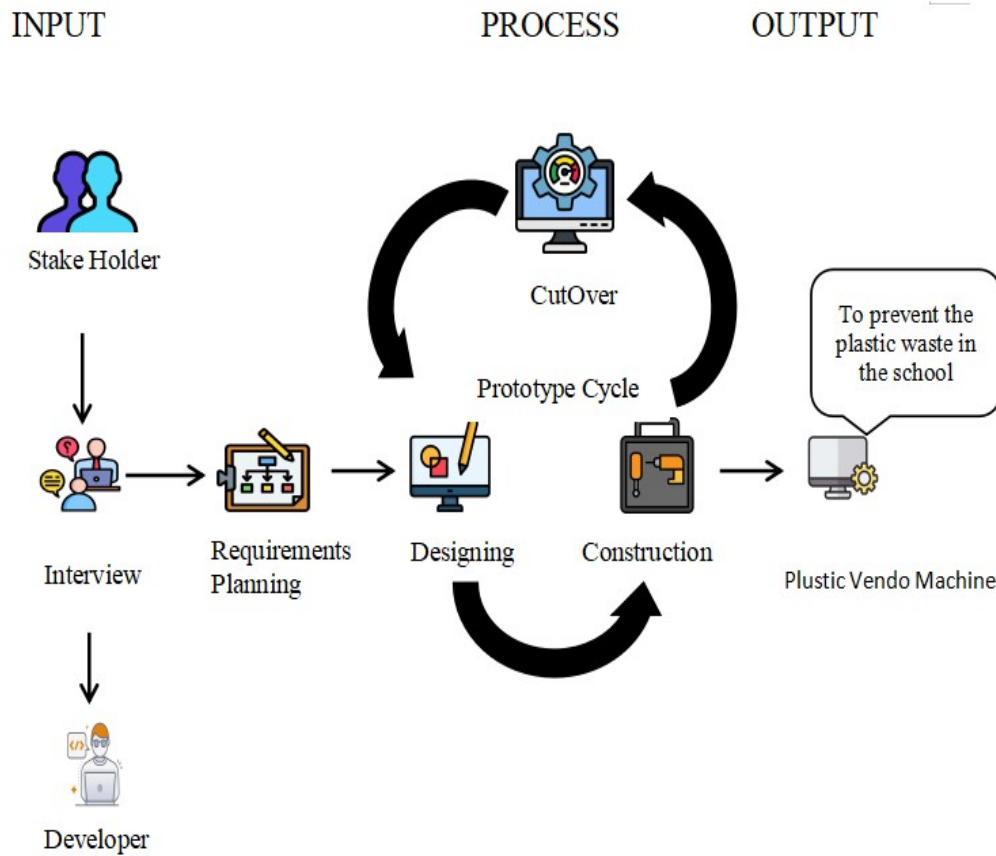


Figure 1.1. Research Paradigm



Statement of the Objectives

This study aims to develop and evaluate the Plustic Vendo Machine, an Arduino-based RVM designed for PET bottle collection inside Golden West Colleges to support proper plastic waste disposal and encourage student participation through a reward-based system. Specifically, the study seeks to:

1. Identify the existing process of collecting PET bottles in Golden West Colleges;
2. Describe the features of the proposed system; and
3. Test the usability of the proposed system.

Significance of the Study

This research is important for Students and Faculty Staff who have an interest in it and directly or indirectly get benefits from the conception and realization of the proposed Plustic Vendo Machine within the school premises.

Students. The Plustic Vendo Machine offers a reward for the students, an easy and engaging means of disposing of plastic bottles in an eco-friendly manner. Rather than just discarding them, students can put plastic bottles into the machine, and they get to exchange valuable items in the claiming station. This practice is beneficial in environmental awareness, proper waste segregation, and the initiative of students to be involved in the recycling process. Students, being promoters of sustainability, also gain the experience of environmental responsibility associated with discipline within their school community, which becomes a positive cycle.

Faculty members. To staff and teachers, the System is a tangible instrument to facilitate environmental education and student involvement in eco-friendly activities. They can bring it in as a topic or program related to environmental science, sustainability, or community service in the school curriculum. Faculty members are able to utilize the device-



generated information to identify students' participation levels and, therefore, include recycling events as a part of school engagements that promote an eco-friendly campus.

Administrative Staff. The Plastic Vendo Machine aids in reducing plastic waste that litters the school area, and thus, by making waste management more facilitative, it benefits the administrative staff. It does away with the challenge of janitors and maintenance personnel by encouraging students to put their plastics in the right places. Moreover, the data from the discarded bottles can be a useful tool for the staff to measure the recycling progress of the school and schedule future sustainability programs.

School Management. The Plastic Vendo Machine is an environmental innovation that school administrators can leverage towards the realization of the school's goals and community values. The measurable outcomes from this are evidence of the school's commitment to sustainability and responsible waste management. Besides, this move turns the school into a green institution and a leader in going digital with green technology and being a model of environmental leadership within the academic field.

Future Researchers. In addition, the present work is a stepping stone for later researchers and students to come up with a newer version of similar environmental technologies or ones that surpass them in effectiveness. The layout, features, and the implementation stage of the system can be a compass to those who aspire to devise innovative solutions to reduce waste and achieving sustainability goals. Moreover, it pays attention to the fact that solving such problems requires integration of engineering, technology, and environmental advocacy rather than dealing with them separately, even in a school-based context.

Scope and Delimitation



The scope of this system covers the design and build of the Plastic Vendo Machine, an automated system that detects and sorts PET bottles with the help of an Arduino Mega microcontroller. The Vendo detects and separates PET (plastic) bottles from non-PET materials using the ESP32 CAM, sensors, servo motors, an LCD display, and a thermal printer. It uses a servo motor to open and close the door and a stopper to completely stop the PET Bottle and Non-PET Bottle from falling in the storage, an ultrasonic sensor to check if the storage is full, and a real-time clock (RTC) to record the date and time of the transactions. The LCD display is used for showing the important system messages, such as “Waiting for PET Bottle” or “Storage Full,” while the buzzer and LEDs give the sound and light indicators of the operations. The thermal printer prints a receipt that shows the total number of PET bottles collected, their points, and the date and time of the transaction. Additionally, the system has buttons for starting the process, opening and closing the door, and answering “YES” or “NO” to prompts when continuation of recycling is asked for. Such a system's primary goal is to offer users a method of automatically recycling PET bottles whereby they first get sorted, then a small receipt acknowledging collection is issued. Through the combination of automation and user interaction, it focuses on increasing awareness of recycling. The device is also intended to deliver a regulated method whereby the door must be closed before the operation for safety and correct system flow. Furthermore, this Plastic Vendo Machine has limitations. The Plastic Vendo Machine is limited itself only to the detection of PET bottles, and as a result, it may not identify other types of plastics or materials. To identify the objects, it uses a pre-trained camera module (ESP32-CAM), which implies that the module may fail under poor lighting conditions or if the bottles are damaged. The system stores a limited number of bottles depending on the storage and does not have an automatic compactor or crusher. Also, the Vendo requires a



manual refilling of the thermal paper used in the printer and manual emptying of the storage area when full.

Additionally, the Vendo is not connected to the Wifi, app or any external database, and the data are not saved digitally or uploaded online. The Vendo has no weight sensor means that the device cannot weigh the bottles or use weight as a factor for detection. It will depend on stable electrical power and a backup battery (UPS), as there is no solar backup included. Furthermore, the time and date on the RTC module have to be set manually, and if the device is turned off for an extended period, it might require resetting. Lastly, this project is only about the demonstration of the working principle of automated PET detection and recycling in a school and is not meant to be used for large-scale or commercial recycling centers.

Definition of Terms

AI (Artificial Intelligence) Image Detection. A technology that enables computers or machines to recognize and classify objects from pictures.

Anti-Fraud Mechanism. A security feature that prevents incorrect or non-plastic items from being accepted by the machine.

Arduino Mega. An electronic board used to control the operation and processes of the vending machine.

DC-DC Buck Converter. A device used to safely regulate and lower voltage for electronic components within the machine.'

ESP32-CAM. A compact camera module used to capture images and identify objects automatically.

PET Bottles Plastic. bottles made from Polyethylene Terephthalate (PET), typically used for water, soda, and beverages.



RAD (Rapid Application Development). A quick and flexible method used to design and build software systems through multiple prototypes and user feedback.

Reverse Vending Machine (RVM). A machine where people can insert used plastic bottles in exchange for rewards like receipts or points.

Servo Motor. A small motor that rotates or moves parts of the machine precisely, used for sorting bottles.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of related literature that serves as the foundation of the study. It examines existing practices in plastic bottle collection, the features of reverse vending machines, and usability considerations relevant to the proposed system. The review aims to identify research gaps and establish a theoretical framework to guide the development and evaluation of the Plustic Vendo Machine.

Existing process in Reversed Vendo Machine

According to Soni et al. (2020), the world is in a dangerous position today as a result of the spread of the COVID-19 pandemic that touched the lives of human beings both physically and psychologically. The pandemic triggered widespread impacts on society, where economic operations like transportation, tourism, and the petroleum sector came to a standstill in most of the world. Throughout this time, over 20 million have been impacted worldwide. Another secondary effect of the pandemic has been an increased demand for single-use plastic bottles, resulting in colossal plastic waste. To counteract this increasingly prevalent issue, there has been increased awareness around proper waste management. Taking advantage of the challenges, researchers came up with a new system to collect and store discarded beverage cans and containers and to decide rewards for customers who participate in recycling. The system utilizes an Arduino-based microcontroller, capacitive proximity sensors, and motors. The proximity sensors are installed at locations adjacent to sides of a conveyor belt to sense incoming bottles. Sensors are wired into a control system, interfaced through the microcontroller, to turn on servo motors used for bottle counting.



Since the system can sense bottles, it can compute rewards to be given back to the consumer, thus promoting correct disposal and preventing the accumulation of plastic wastes and undue expenditure problems. Programs like the Smart Cities Initiative and the Swachh Bharat Abhiyan are only a few of the government programs enabling such programs

According to Wong et al. (2019), the 3Rs, or Reduce, Reuse, and Recycle, are accepted principles in dealing with the ever-increasing fallout from waste. With limited space for landfills, recycling takes the form of primary waste management. One primary drawback of conventional recycling is that it oftentimes requires individuals to carry their recyclable waste in bulk to a recycling center, which is inconvenient and discouragingly less participative. To these ends, the reverse vending machine was developed as an autonomous vehicle with its built-in reward system for recycling. The RVM is based on an Arduino microcontroller comprised of various sensors used to detect certain recyclable units. The user is rewarded for correctly identifying a metal item or a plastics bottle within the machine, helping produce repeat use and higher reliability of waste disposal. Finally, this could also be one viable way of poverty alleviation through a waste-to-wealth model.

According to Yaddanapudi et al. (2023), plastic pollution, especially from water bottles, is a growing concern worldwide. Only recycles 10% of plastic bottles; the rest are either crushed in landfills, thrown into open drains, or tossed into the oceans, with devastating consequences to marine life. It's reported that one plastic bottle is purchased every second, equating to nearly one million bottles every minute and a predicted increase of 20% by 2021. Indirect consumption of plastics through the food chain is an additional health hazard. Against this background, the present work is aimed at achieving 100% recycling of plastic bottles by integrating the entire population into the recycling process.



The concept proposes recognizing and sorting plastic bottles automatically using a Haar Cascade classifier in the RVM prototype.

According to Alfarisi et al. (2022), various factors including atmospheric temperature, air saturation, and illumination intensity have bearings on the RVM sensor accuracies. User interaction such as inserting containers also influences accuracy. To remove these challenges, the researchers designed a sensor scanning area that reduces disturbances as well as guides users to insert proper containers. The system combines an Arduino UNO microcontroller with the scanning sensors, conveyor motors, dynamo motors as drive systems, and linear actuators for pressing. Servo motors manage the coin dispenser. The main purpose of this RVM is segregation of plastic from other types of containers and cans for appropriate storage. Data collected from the sensors are classified and analyzed to ensure accurate processing. Scanning sometimes does not seem to work perfectly, yet the rest of the performance remains that one can still get a container released to the user, even though there are some incidences in rare cases where misclassification exists.

According to Dacay et al. (2020), the Philippines are facing rising challenges in waste management, with an average solid waste generation of 35,580 tons a day. In urban areas, household-generation waste is about 0.5 kg/day, while in rural areas it is approximately 0.3 kg/day. This problem encouraged the creation of the VendoBin—a waste bin and vending machine hybrid—to encourage the proper disposal of plastic bottles. The system grants users usable items, such as ballpoint pens or bundles of newsprint, in exchange for depositing plastic bottles. The VendoBin prototype is being controlled through a RaspBerry Pi 3 single-chip computer. Infrared sensors are used to classify plastic and non-plastic items while the ultrasonic sensors check the fill level of the bin. Points earned through each transaction were recorded in a text file database. Users receive a unique code for redeeming their reward items, while automatic notifications will alert



authorized services when the bin is ready for emptying. Evaluation showed that the VendoBin was able to successfully classify 100 of the items tested, along with smooth code generation for both new and returning users for reward-issuing purposes.

According to Cervantes et al. (2021) plastic waste in the Philippines is poorly handled. Most of the discards of PETE (Polyethylene Terephthalate) bottles filled with water are expected to end up in water bodies. With this, the researchers suggested a shredder system incorporated into RVMs. This system is to reduce the solid wastes for more efficient storage and recycling. An automated cap separation stage is included such that it will separate caps from bottles depending on the materials. Tests conducted using PETE bottles from 300 to 500 mL demonstrated that volume reduction after shredding is about 50–65%. Bottles which are between 300 and 400 mL size have a volumetric shredding success rate of roughly 92%, whereas the cap detection system could also get 88% accuracy. Thus, the results proved that the system readies the PETE bottles for recycling within the operations of RVM.

According to Rubio and Lazaro (2016), the aim of the solar-powered Reverse Trash Vending Machine developed is to motivate a more active participation in recycling and thus reduce irresponsible waste disposal in the Philippines. The system uses a Gizduino X ATmega 1281 as the main processor, aided by a Gizduino 644 microcontroller along with a GSM communication module. The programming of the device was done on an Arduino IDE, version 1.6. The machine uses a 15V rechargeable battery for power, which is constantly recharged by a solar panel mounted on it, thus promoting sustainable energy. The system accepts empty plastic bottles with a capacity of up to 500 mL and a base diameter of 3.5 inches, as well as aluminum cans. The materials will then be scanned, crushed, and stored in a holding bin with the type of material determined using an inductive



proximity switch. Incentives are provided in the form of cash rewards, aimed at motivating users to recycle.

According to Dumpayan et al. (2017), the development of an RVM has been undertaken to promote responsible waste management at Colegio de San Juan de Letran. This system is based on a microcontroller that will take care of input and output devices: sensors, keypad, display, and motors. Plastic bottles deposited by the user are converted into points, which are redeemable for items stored in the machine. The RVM will read the RFID account and accurately differentiate between plastic from other non-plastic items; it then stores and updates points, and dispenses rewards. The unit can be operated via solar or commercial power, wherein the solar panel and battery serve as backup in case of power interruptions. Survey results show favorable responses from students, which indicates that the presence of the RVM could largely affect waste segregation in school.

According to Hakim et. al., (2020) described a reverse vending machine arrangement whereby customers are rewarded with cash-for collecting empty bottles. Several techniques are used for the acceptance and sorting of bottles; these include material identification through infrared spectrometry, barcode reading, and image processing for bottle shape recognition. These technologies ensure correct assignment of recyclable containers and subsequent rewarding of users.

According to Tomari et al. (2019) studied several challenges in the misuse of recycling bins found in schools, universities, shopping complexes, and residential areas. Even with awareness campaigns, prohibited materials would find their way into the bins. To overcome this, the researchers made an RVM with a material identification module to accept only empty recyclable materials. The system revolves in various mechanisms with the help of sensors detecting PET bottles and aluminum cans. In the tests, the module rejected unwanted materials with an average accuracy of 82.5%, thus assuring that



recycling efforts would stay useful while giving users incentive through some form of reward.

The widespread use of single-use PET bottles has led to increasing plastic pollution, particularly in the Philippines, where much of the waste ends up in waterways or landfills, creating environmental and health hazards (Soni et al., 2020; Cervantes et al., 2021). Traditional collection methods rely on manual segregation and user compliance, which are often inconsistent and inconvenient, limiting participation (Wong et al., 2019). Automated systems like Reverse Vending Machines (RVMs) address these challenges by using sensors and microcontrollers to detect, sort, and store PET bottles while providing rewards to encourage proper disposal (Alfarisi et al., 2022; Dumpayan et al., 2017). These systems enhance the efficiency and reliability of PET bottle collection, ensuring that more materials are correctly processed for recycling (Cervantes et al., 2021). Despite occasional sensor inaccuracies, RVMs offer a more reliable and user-friendly approach, enhancing recycling participation and supporting sustainable waste management.

Features of Reversed Vendo Machine

According to Tahan & Arslan (2021), perhaps the most striking characteristic of a reverse vending machine is that it encourages recycling and minimizes harm to the environment. Human activities such as littering beverage packaging have wreaked havoc on the natural world. Because wastes decompose extremely slowly, recycling became an imperative. The reverse vending machine is designed to integrate recycling into individuals' day-to-day activities. It makes the customers put in empty bottles or plastic cans and rewards them, normally by card payments or points. It is very small in size so that it can



easily be mounted in schools, malls, markets, and offices. It is employed to advocate for recycling awareness and encourages individuals to engage in environmental preservation.

According to Zia et al. (2022) one of the key aspects of a reverse vending machine is that it can offer a system of incentive-based recycling. Due to the increasing amount of used plastic bottles clogging landfills and waste management facilities, recycling has been a strategic decision to offer environmental protection. The reverse vending machine provides a cutting-edge, portable way of incentivizing individuals to recycle for rewards on plastic trash. Besides reducing the environmental impact, the proposal also generates economic opportunity with recycling materials. Additionally, the machines' design targets long-term lifespan, effectiveness, and applicability to real-world use, thus presenting an effective solution to plastic waste issues.

According to Kim et al. (2021) the reverse vending machine is an automated waste sorting and collecting system intended to enhance recycling efficiency. Owing to the fact that plastics are used abundantly and produce vast quantities of waste, there must be appropriate management. The proposed RVM is a mini automatic sorter that can differentiate recyclable wastes like plastics, glass, and aluminum cans. It acts in three main steps—collected, identification, and sorting the waste. The machine recorded a commendable sorting rate of about 95% and is thus suitable for small-scale environments such as convenience stores. The attribute enables efficient and fast on-site segregation of the waste, reducing the burden on massive recycling plants and promoting systematic recycling behavior.

According to Juansah et al. (2024) that a reverse vending machine is made to reward users for returning used plastic bottles and other waste materials, one at a time. The system is configured on different sensors like the load cell (HX711) for weighing measurement,



light-dependent resistor (LDR) for detecting light, and ultrasonic sensor (HCSR04) for measuring distance. All these components are governed by an Arduino Mega 2560 microcontroller, which is integrated to provide efficient detection. In the study, it was found that the system efficiently detected plastic bottles of various sizes with a detection accuracy of 95.33%, using various trials. This operation shows the accuracy and reliability of the machine to detect and process recyclable material correctly.

According to Simatupang (2022) a reverse vending machine for waste management which sorts and recognizes recyclable cans and bottles automatically. It employs a TCRT5000 sensor to recognize bottles and an inductive proximity sensor to recognize metal cans. It directs every item detected to its respective bin, and the machine computes rewards based on the items inserted in total. A push button can be used to let customers print a reward ticket, provided that only recyclable materials are sensed. The RVM is equipped with ultrasonic sensors to check whether the bins are full, and LED lights indicate the machine's status. The system successfully performed 24 tests at 100% accuracy, demonstrating that it is reliable and accurate in sensing and processing recyclable waste.

According to Lopez (2021) focused on the use of image processing technology in reverse vending machines to identify and recycle polyethylene terephthalate (PET) bottles. Since many PET bottles remain unrecycled, the study aimed to improve sorting accuracy using the YOLOv3 algorithm, known for its high speed and precision in object detection. The system uses a camera to capture an image of the bottle, which is then analyzed by a Raspberry Pi to classify whether it is recyclable. Once accepted, the machine activates a DC motor to open the PET bottle compartment, and the bottle type and processing time are displayed on an LCD screen. The experiment achieved a 90% accuracy rate, showing that reverse vending machines with image recognition can effectively support PET bottle recycling.



According to Yoo et. al. (2022) a reverse vending machine called Clean Bottle, which would aid South Korea in implementing the policy of sorting transparent PET bottles for high-quality recycling. The machine is fitted with image recognition and convolutional neural network (CNN) to identify if an inserted bottle is clean, empty, and does not have a seal. The system is run on Raspberry Pi 4B and PiCamera to facilitate efficient processing and classification. The machine was over 99% correct during testing in the identification of acceptable PET bottles. The feature helps collection occur on solely properly prepared bottles, which enhances material quality overall and assists the government recycling efforts.

According to Colaljo et al. (2024) Population growth and urbanization create increasing waste amounts made up of plastic bottles and canned beverages, thus making proper waste management more relevant than ever. This research developed an image-based reverse vending machine that takes recycling as a means of public participation via reward systems. The RVM employs a Raspberry Pi coupled with YOLOv5 image detection technology for precise identification of plastic bottles and aluminum cans. A load cell sensor checks for the emptiness and weight of the items before they are processed. Highly accurate and reliable, the system has performed excellently with 96% recall, 97% precision, and 96% F-measure. In addition to offering low-cost upgrades for the whole system via image detection, this setup is therefore fully scalable and future-proofed to make it easily applicable in further waste management development.

According to Handoko et al. (2018), the reverse vending machine is an innovative mechanism for accepting used bottles and rewarding users with cash or tickets in return. This is a technological innovation that exploits artificial intelligence at identifying and classifying different bottle sizes-that is, whether 300ml, 600ml, or 1500ml. It automatically issues a given number of tickets that can be exchanged into cash, depending on the bottle



size. The machine is mainly intended to collect used plastic mineral water bottles and is said to have been developed through experimentation and testing based on R&D. Building this model aims to keep simplicity and low production costs, making it practical and affordable. Importantly, it is also one of the first uses of such a vending machine in Indonesia for exchanging plastic bottle waste.

According to Dhulekar et al. (2018), Bottle Recycling Machines (BRMs), or reverse vending machines, are essential for curbing solid waste pollution due to discarded bottles. Their study proposed an idea where the used bottles are automatically collected and identified and sorted using machine learning algorithms. The detection of bottles and guidance of users through the recycling process is initiated by the motion of the camera attached to a Raspberry Pi. After detecting the bottle, the machine prints a coupon through a thermal receipt printer as a reward. The bottles are sorted on the basis of material type through an array of counterweights and are sent for recycling. The overall system works on a low-cost Python-based machine learning program, ensuring they identify and sort bottles effectively and economically for waste management.

Reverse Vending Machines (RVMs) are designed to promote recycling by automating the detection, sorting, and storage of PET bottles and other recyclables while providing user incentives (Tahan & Arslan, 2021; Zia et al., 2022). They employ a combination of sensors, including load cells, light-dependent resistors, ultrasonic sensors, and inductive proximity sensors, to accurately identify materials, with detection rates reported between 90% and over 99% (Juansah et al., 2024; Simatupang, 2022; Lopez, 2021; Yoo et al., 2022; Colaljo et al., 2024). Many RVMs incorporate image recognition and machine learning algorithms, such as YOLO and convolutional neural networks, to further enhance sorting accuracy and ensure only suitable materials are accepted (Lopez, 2021; Yoo et al., 2022; Colaljo et al., 2024). To encourage consistent participation, these



machines provide rewards—points, cash, or redeemable tickets—automatically computed based on the items inserted (Tahan & Arslan, 2021; Handoko et al., 2018; Dhulekar et al., 2018). Designed for portability and ease of use, RVMs can be deployed in schools, offices, malls, and convenience stores, and are powered by accessible microcontroller systems like Arduino or Raspberry Pi. Collectively, these features make RVMs an effective, reliable, and user-friendly tool for enhancing recycling and supporting sustainable waste management.

Usability of Reversed Vendo Machine

According to Jungthawan et al. (2024) reverse vending machines (RVMs) promote recycling to reduce plastic waste, specifically PET bottles, in Thailand. The research found only around 4.5% of all plastics are actually getting properly recycled, the remaining amount goes off spreading microplastics into air and environment. The RVM programmed provides incentives for rewarding and thus encourages people to return the bottles. However, the study of data from the 15 RVM units that were installed in hypermarkets from 2020 to 2022 reveals that an average of an abysmal 670 kg of PET bottles was collected per RVM annually rather than the ideal target of 3200 kg. The study also revealed that RVMs helped in raising awareness of recycling but may not yet make an economic case. Future improvements in the business model and government policies will need to effectively pave the path for practical and sustainable RVMs in the longer term.

According to Pramita et al. (2019) the Reverse Vending Machine (RVM) could become a very effective tool to control plastic waste in India as solid waste is estimated at 80–85 million tons by 2030. The RVM accepts barcoded plastic bottles and provides money or redeemable coupons in return, thus making recycling easier and more rewarding.



Nevertheless, the study found that use of RVMs in the country is still limited. To find out the reason, the researchers developed a conceptual model that identified key factors influencing usability and adoption: awareness, willingness to participate, incentive availability, convenience of use, and user involvement. According to the study, these factors deserve enhancement so that public acceptance and habitual practice of RVMs can be improved so as to achieve eco-friendly waste management.

According to Shamsi (2021), research on Reverse Vending Machines (RVM) was conducted so that Dubai could increase more utilization on the part of technology towards environmental conservation and sustainability. One of the areas covered by the research was a significant reduction in plastic waste (especially plastic bags), the introduction of RVMs in Union-Coop supermarkets (one of the largest retailers in the city), which all signified the concept behind Dubai using up-to-date technology for enhancing citizens' quality of life while preserving the environment from excessive depletion of resources. The study points out that the benefit of adopting RVMs is justified in urban areas such as Dubai, in terms of engendering both consumer convenience and environmental accountability towards efficient and easier recycling.

According Nurfikri & Martono (2023) to Plastic bottles slowly biodegrade naturally. Indonesia releases plastic bottle trash uncontrolled and can harm the health of people and the environment. Indonesia began to digitalize plastic bottle recycling using a circular economy concept as a solution to solve this issue. Plasticpay is a social movement rooted in the cyber world that encourages people to recycle plastic rubbish, which first harms the environment, into economically recoverable material. Plasticpay has partnered with various stakeholders such as the University of Indonesia Vocational Education Program to recycle plastic waste. Yet field study reveals that the Reverse Vending Machine is not utilized on a mass scale. This research analyzes obstacles to adoption and intention to



use Reverse Vending Machines. Known variables are subjective norms, attitude, and behavioral control. The questionnaire was drafted, and opinions were collected from 80 participants in order to know their views. Structural Equation Modeling is applied to identify the conceptual model. Results from the company show that the factors of behavioral control impact the usage of Reverse Vending Machines. Practicality in having a big Reverse Vending machine demands ease of use and incentives.

According to Amantayeva et al. (2021) undertook systems engineering research to investigate how reverse vending machines could optimally be incorporated into Kazakhstan's waste management system. The research focused on public awareness, recycling habits, and acceptance of RVMs in order to explore the reasons for the country's very low recycling rate. Surveys identified user needs and motivations for an improvement RVM model via the development of a House of Quality (HoQ) and Functional Flow Block Diagram (FFBD). Besides, it considered a risk assessment of a national roll-out of RVMs, including intervention measures for reducing the difficulties encountered with their introduction. This work shows that understanding user behavior, incentives, and reliability are primary ingredients for enhancing usability and acceptance with respect to RVMs in public waste management.

According to Tiyyarattanachai et al. (2015) discusses the potential uses of RVMs in enhancing the efficiencies of recycling programs and the qualities of recycling feeds, especially for PET, one of the most popular beverage packaging materials. RVMs find good applications in countries like Germany and the USA, clearly establishing their reverse logistics systems for the direct collection of recyclable materials from the hand of the consumer. In Thailand, where such machines are more recently introduced, the study examined consumer acceptance and the influence of RVMs on recycling performance. Results indicated that in the areas to which RVMs were introduced, the collected amount of



PET bottles was higher by 21% compared to sites that utilized only traditional waste collection methods. Moreover, bottles collected through the RVMs were deemed cleaner, as they were remote from contamination by caps, straws, and non-recyclable materials. This indicates that RVMs could then be seen as efficient tools that enhance qualitatively and quantitatively the paradigm of recycling facilities.

According to Dulaca et al. (2025) Many people today seek the heroic intervention of a scientific lightning bolt to absolve humanity of plastic pollution. However, worked on plastic bottle pollution in Muzon Proper, where participation by the youth in recycling is still very low. The researchers adopted an innovation called MuzonEcoSave, through which a Reverse Vending Machine (RVM) was set up to enhance user participation through the combination of advanced detection technologies and reward incentives. The RVM employs simultaneous loading with load cells and sensory detection through the use of inductive sensors and infrared technology to ensure the accurate and economical classification of plastic bottle waste. Overall, it was able to perform with 97.14% accuracy in detecting plastic bottles in less than 7.56 seconds per bottle. The cloud-based reward-based feature, in fact, paved way for a 94% increase in youth participation when compared with the traditional methods. Ultimately, the study showed that coupling accurate detection machinery with motivational incentive system greatly enhances the usefulness and efficacy of RVM systems in stimulating recycling behavior of the community.

According to Clorado et al. (2025) the major and chronic problem of polyethylene terephthalate (PET) bottle pollution in the Philippines owing to their limited and low environmental awareness recycling systems. The research provided AI-based Reverse Vending Machines (RVMs) that had optical reader technology for improved collection of PET bottles and recycling in universities. It was conducted in the National Capital Region (NCR) through both survey and experimental methodologies to determine the usability and



effectiveness of the system. From a total of 408 respondents—mostly students aged 18-24—results displayed marked engagement and positive user experience. Statistical analysis revealed that machine learning algorithms and a stable power supply are the key factors when resulting in system performance. Experimental trials showed that the machine has 92% accuracy in recognizing bottles without lids and labels, and allows each and every user to process up to three bottles in a session. Beyond that, the trials reveal very much increased collection speed and precision, indicating that AI-based RVMs can productively upgrade recycling operations as well as motivate sustainable behavior through reward-based motivation.

According to Tur et al. (2019) Most commonly, researchers have either already studied or are still studying the RVM technology in their work. Reverse Vending Machines, or RVMs refer to automated systems enabling consumers to return used beverage bottles and cans for recycling. Whereas in the Russian Federation, large-scale adoption of RVMs is hindered by gaps and inconsistencies in the legislation on producers' extended responsibility concerning waste management. The research indicated that automating RVMs can recycle plastic and metal waste for private consumers, supplemented by a consumer-incentive motivational scheme including discounts and bonuses in mainstream retail establishments. Standard camera technology serves as the initial criteria for the identification of the materials marketed. Thus, it lowers the operational costs while further sorting will take place at specialized workshops before being transported to recycling facilities. From there, it could be transformed into materials used in producing machine components and packaging materials, which include aluminum frames and plastic elements processed through 3D printing. In the end, the findings of the research indicate that such an approach would increase economic efficiency and ease of using RVMs as recycling systems significantly.



According to Lohani et al. (2023) find that billions of plastic bottles and aluminum cans are consumed daily throughout the world, with most of them ending up in landfills or being scattered as litter. Even though recycling of such materials can transform them into reusable household products, it is often found that the cost of recycling plastic bottles exceeds that of producing new ones, thus rendering it less appealing economically. Effective segregation has been recognized as one of the main challenges for recycling, because manual separation of recyclable items is labour-intensive and decreases efficiency. As a solution, the researchers have proposed an automated system that would use the application of machine vision and deep learning (CNN) to identify and separate the plastic bottles and aluminum cans accurately. Different from conventional sensors, which lack precision, the AI-enabled system is capable of classifying waste items with a 96% accuracy and automatically sorting them into particular compartments for plastics or cans, which receives a recyclable code as a reward. The research concludes that the automation of recycling through reverse vending technology can thus become more economic, efficient, and user-friendly, increasing the sustainability of waste management in general.

The usability of Reverse Vending Machines (RVMs) plays a critical role in encouraging PET bottle recycling, as user participation is influenced by ease of use, accessibility, reliability, and clear reward mechanisms (SK et al., 2019; Amantayeva et al., 2021; Dulaca et al., 2025). Studies from Thailand, India, Indonesia, the Philippines, and other countries indicate that while RVMs can raise awareness and improve recycling quality, actual usage often falls short of targets due to limited public knowledge, convenience issues, and socio-economic barriers (Jungthawan et al., 2024; Nurfikri & Martono, 2023; Tur et al., 2019). Integration of AI, machine learning, and image recognition enhances detection accuracy and speeds up processing, further improving user experience (Clorado et al., 2025; Lohani et al., 2023). Research shows that when RVMs are



designed with intuitive interfaces and effective incentive systems, they significantly increase both participation and recycling efficiency, suggesting that usability is a key factor in achieving sustainable waste management outcomes.

CHAPTER 3

METHODOLOGY

Research design

This study used an experimental-descriptive research design to develop and evaluate the Plustic Vendo Machine. The experimental aspect involved designing, building, and testing the machine to ensure it performed according to its intended operation. The descriptive aspect involved collecting user feedback to assess the system's usability among students at Golden West Colleges.

The developers followed the Rapid Application Development (RAD) model, which includes four stages: (1) Requirements Planning, where project needs and goals were defined; (2) User Design, where prototypes were created; (3) Construction, where hardware and software components were assembled and tested; and (4) Cutover, where the completed machine was deployed for usability evaluation.

To assess usability, the developers conducted a survey-based test involving fifty student participants who used the Plustic Vendo Machine in an actual setting. After operating the system, participants completed a structured Likert-scale questionnaire measuring ease of use, clarity of instructions, responsiveness, and overall experience. The data were analyzed using descriptive statistics, specifically frequency and percentage, to summarize the results and identify areas for refinement.

Overall, the research design integrated prototype experimentation and usability evaluation to examine how an automated vending system could support proper PET bottle disposal and recycling practices in the school.

Software Development Model

Rapid Application Development (RAD) is not a recent concept. It has nonetheless existed since the 80's. It does not fall under a singular model like the waterfall model does. It is an ever changing development foundation based on the needs and technologies related to application development at that given moment in time. (Kissflow, T. 2025, April 30).

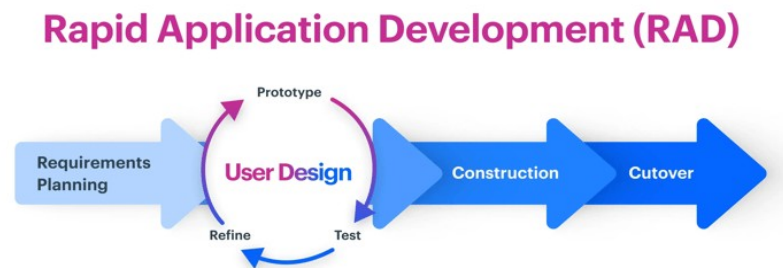


Figure 3.1. RAD Life Cycle

The Stages of Rapid Throwaway Prototyping Model

Requirements Planning. The beginning of this stage to discuss the outcome of project necessary needs, scope, and to implement the ideas into the actual project.

In this stage, the developers will test their critical thinking to achieve the possible outcome of this project, and to use the potential of the Arduino to make the Reversed Vendo Machine, to reduce the plastic waste.

User Design. The objective of this stage is to create a framework of the Plustic Vendo Machine and to see the possible design, flow of the machine, and the adjustments of the actual Machine.

The developers will going to the trial and error to build a unfinished machine. The developers will seek the best model, version of the project, and to test the durability of the frame.



Construction. This stage will test the machine to debug the errors, to spot the missing features of the machine, and to prevent the possible malfunction in this project.

The developers will adjust the flow of the machine based on the reviews and feedbacks of the clients.

Cutover. This stage will be the final for the actual project. After the planning, testing and prototyping the machine will finalize the actual model.

The developers deploy the machine in the target area or the the target client, this machine will help the are to become responsible for the enviroment and collect a reasonable prize, and to become eco-friendly, efficient in the environment.

Participants/Respondents

In this study, the researchers gathered data primarily through survey questionnaires and online research. A total of 50 respondents participated in the survey, consisting of students who frequently use the Plastic Vendo Machine within the school premises. The questionnaire served as the main instrument for collecting relevant data, allowing the researchers to understand the participants' experiences, opinions, and suggestions regarding the system. To ensure clarity and accuracy in the responses, the researchers provided necessary guidance and explanations while the respondents completed the survey. This approach helped participants fully comprehend each question, resulting in more meaningful and reliable answers.

In addition to the survey, online research was conducted to supplement and validate the data collected from respondents. The researchers explored credible internet sources, including articles, research studies, and technical reports, to obtain additional insights into similar Plastic Vendo Machine projects, effective recycling practices, and technical information related to the system's components. Only up-to-date and trustworthy sources



were used to ensure the accuracy and credibility of the gathered information. By combining primary data from survey questionnaires with secondary data from reputable online resources, the researchers were able to obtain a comprehensive and well-rounded understanding of the study. This multi-source approach ensured the depth, reliability, and validity of the findings, thereby strengthening the overall quality and credibility of the research.

Research Instruments

The data that the developers collected is covered in this section. a research component to guarantee the study's accuracy by measuring and analyzing the data.

Survey Questionnaire. A questionnaire is a research instrument consisting of a series of questions designed to collect data from respondents. It is widely used in surveys, academic research, market studies, and evaluations to gather information about individuals' thoughts, preferences, behaviors, or experiences. Questionnaires are an essential tool in quantitative research, helping researchers gather standardized data from large groups of people.

Online Research. Online research comprises the process of information gathering using internet materials such as articles, studies, and even reports. During the course of the project, the developers resorted to online research to gather information pertinent to other RVM projects, appropriate practices in plastic recycling, and technical data regarding the parts of the machine. Such a study was useful in determining the most common errors that were associated with the RVM and how best the challenges could be addressed. All the information collected through interviews and other online materials would provide sufficient data for improving the RVM by making it more efficient.



Tools for Data Analysis

Flowchart. A flowchart is a visual representation of the data that shows a certain process or the flow of instructions of an algorithm which is essentially a step-wise method of solving a bigger task. Typically, a flowchart diagram depicts the different steps as rectangles and the mode of interaction or linking between the rectangles is done by arrows. By this diagrammatic method, a flowchart exhibits the entire model of a certain problem and arrives at the solution by the correct method. It finds its application in programming which includes program analysis and design as well as in other sectors.

Gantt Chart. A Gantt chart is a type of horizontal bar chart that illustrates the timeline of a project. Basically, each task is visually depicted by a horizontal bar, where the bar's length corresponds to the task's time frame. The location of the bar on the chart reflects the task's starting and finishing dates.

Requirements Analysis

Recycling should be convenient, enjoyable, and appealing for all, yet traditional recycling methods of plastic bottles make it difficult for users to feel motivated to do so. The general public is compelled to drop bottles off at specific centers, inconveniently located bins, or awaits for funds to be rewards first. It is evident that not everyone is geared to be incentivized towards recycling with great effort through convenience.

Traditional methods make it difficult to feel compassionate towards the endeavor while many want to do their part, not everyone can actively engage without motivation. The individual aims to recycle plastic bottles but struggles to find collection stations near them, which hampers their willingness.

The general public is also forced to manually and endlessly sort bottles for recycling without any direct guarantee. It is clear that the effort put in wont lead to any end results

making the endeavor meaningless. RVMs allow users to do just that. They can drop PET plastic bottles into the system with ease. Through the systems' AI sensors, RVMs check if the bottle inserted is valid and a reward is immediately dispensed on successful validation through receipt paper. Automatic validation of reward systems and effortless grants enable users to feel motivated to complete the task allowing reductions in the struggles associated with plastic bottle recycling.



Figure 3.2. Inserted a bottle to get reward

Tools and Technologies Used

Arduino Mega 2560. The Arduino Mega 2560 is a versatile microcontroller board based on the ATmega2560, designed for complex and large-scale projects. It features 54 digital I/O pins, 16 analog inputs, 4 UARTs (hardware serial ports), and a large program memory. Ideal for projects requiring multiple sensors, actuators, and communication modules, such as robotics, automation, and IoT systems.

FR4 universal protoboard PCB. An FR4 universal protoboard PCB is a sturdy and reusable board made from fiberglass-reinforced epoxy material, known for its strength and heat resistance. It features a grid of pre-drilled holes with copper pads that allow users to solder electronic components and create custom circuits. This makes it ideal for building, testing, and prototyping electronic projects. Unlike regular breadboards, it provides a more



stable and permanent connection through soldering, ensuring better durability and performance.

Videoke Buttons. These are large, durable push buttons similar to those used in karaoke or arcade machines. They serve as user input devices in interactive systems, allowing users to trigger specific functions, such as starting a process, confirming selections, or controlling specific actions in a project.

Jumper Wires. Flexible insulated wires with pin connectors at both ends, used for making quick and reliable electrical connections between components on a breadboard, Arduino, or other prototyping setups. They come in various types, such as male-to-male, female-to-female, and male-to-female, suitable for different pin configurations.

ESP32-CAM. A compact camera module that integrates a 2MP OV2640 camera with an ESP32 chip, offering Wi-Fi and Bluetooth capabilities. It supports image capture, video streaming, and even AI-based image recognition through platforms like Edge Impulse. Widely used in surveillance, facial recognition, and IoT applications.

DC-DC Buck Converter. A voltage regulation device that steps down a higher DC voltage (e.g., 12V to 5V) to a lower, more usable level. It is crucial for maintaining stable voltage in electronic projects where different components require varying voltage levels. Buck converters are efficient and help reduce power loss compared to linear regulators.

LCD Display. A module used to display text, numbers, or graphics. Typically, a 16x2 LCD module with I2C interface is used in Arduino projects to present information like status messages, sensor readings, or user prompts. It enhances user interaction by providing visual feedback.

Servo Motors. Electric motors capable of rotating to a specific angle, typically ranging from 0° to 180°. Controlled by pulse width modulation (PWM), they are commonly



used in robotics for precise position control, such as moving levers, controlling flaps, or creating automated mechanisms.

Thermal Printer. A compact printer that uses heat to transfer ink from a ribbon to thermal paper. Often used in portable receipt printing, it connects to microcontrollers like Arduino via serial communication. Thermal printers are ideal for generating quick, on-demand printouts of text and simple graphics.

UPS (Uninterruptible Power Supply). A device that provides emergency power when the main power source fails. It consists of a battery backup system to keep critical components running during outages, protecting sensitive electronics from sudden power loss or fluctuations.

Resistors. Passive electronic components that limit current flow, essential for controlling voltage and protecting sensitive components. They are used in voltage dividers, signal conditioning, and current limiting applications. Resistors come in various resistance values, measured in ohms (Ω).

Quick Wire Terminal Connector. A simple, fast, and secure way to connect multiple wires without soldering. These connectors feature push-in terminals that hold wires firmly, allowing quick assembly and disassembly. They are ideal for rapid prototyping and maintaining reliable connections in complex wiring setups.

LED (Light Emitting Diode). A semi-conductor device that emits light when current flows through it. LEDs are energy-efficient and widely used as indicators, status signals, or illumination in electronic projects. They come in various colors and sizes, with low power consumption.

5V Passive Buzzer. An audio device that produces a buzzing sound when supplied with a voltage, typically 5V. Unlike active buzzers, which generate sound independently,



passive buzzers require an external square wave signal to produce a tone. Often used for alarms, notifications, or simple sound effects.

Implementation Plan

When the development is complete, the Plustic Vendo Machine will be demonstrate infront of the client. It will deploy the Plustic Vendo Machine after the clients approval and agree in the user documentation. That's a proper deal to have a permission of the client in the developers to maintain the System, to innovate more the Plustic Vendo Machine.

Strategy	Activities	Persons Involve	Duration
Checking the System	Deployment in the school	Researchers, Developers	1 Day
Encourage and Spread the information	Promote and posting online platforms	Researchers, Developers, users, faculty staff, Administrators	8 Months

Table 1. Implementation Plan

Budget Plan

This part highlight the budget allocated for the execution of the Plustic Vendo Machine system.

Table 2 to show the approximated cost involved to initiate and to use the proposed system.

Developer's Fee. This covers the expenses by the developer's during the system development stage. This is base on the estimated expenses of the system, total cost, damage, and other necessary expenses.



Implementation Fee. This involve all expenses from the planning stage to the successful deployment of the machine, components, and other important materials.

Development	Amount
Developers Fee	Php 8,000
Implementation Fee	
Arduino Mega, Esp32 cam with OV2640, Ov5640 5mp camera module	Php 1,426
Inplay Ups 650VA, Senda portable thermal printer, Dc-Dc Buck Converter	Php 2,295
V380 CCTV small Hd Hidden camera, Enclosure Box, Servo Motors, Jumper Wires	Php 1,442.25
Prototype Materials (Jumper wires, Karaoke Buttons, Quck Wire Termail Connector Universal, Xprinter XP-58IID thermal printer)	Php 1,172
Total	Php 23,335,25

Table 2. Budget Plan

CHAPTER 4

RESULTS AND DISCUSSION

The Existing Process

The students deposit their used PET bottles and other recyclable materials in designated bins and metal recycling cages, which the LGU truck collects and delivers to a local recycling partner for proper processing or resale. This method relies on manual collection and sorting to manage recyclable materials. It encourages proper waste segregation, but the absence of an automated collection system or incentive mechanism limits its overall efficiency and long-term sustainability.



Figure 4.1. Existing Process of Collecting PET Bottles



Flowchart

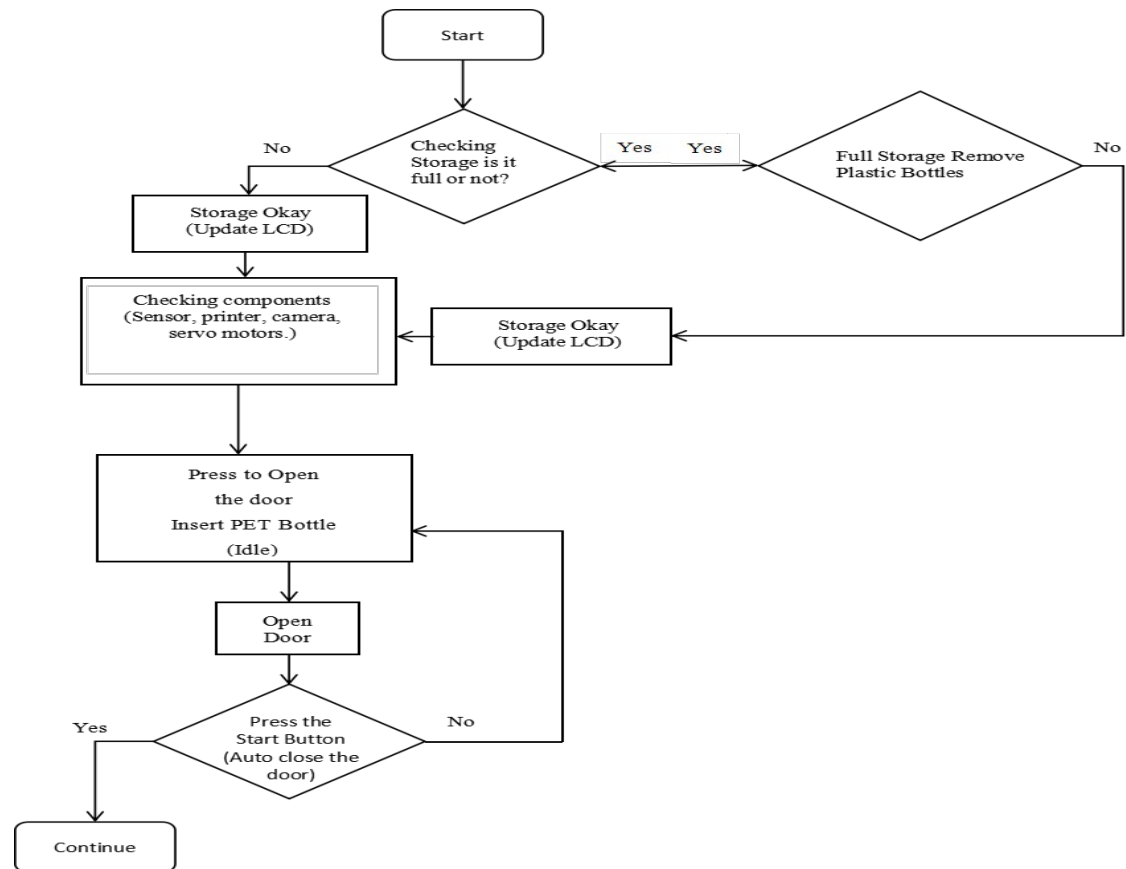


Figure 4.2. Flowchart

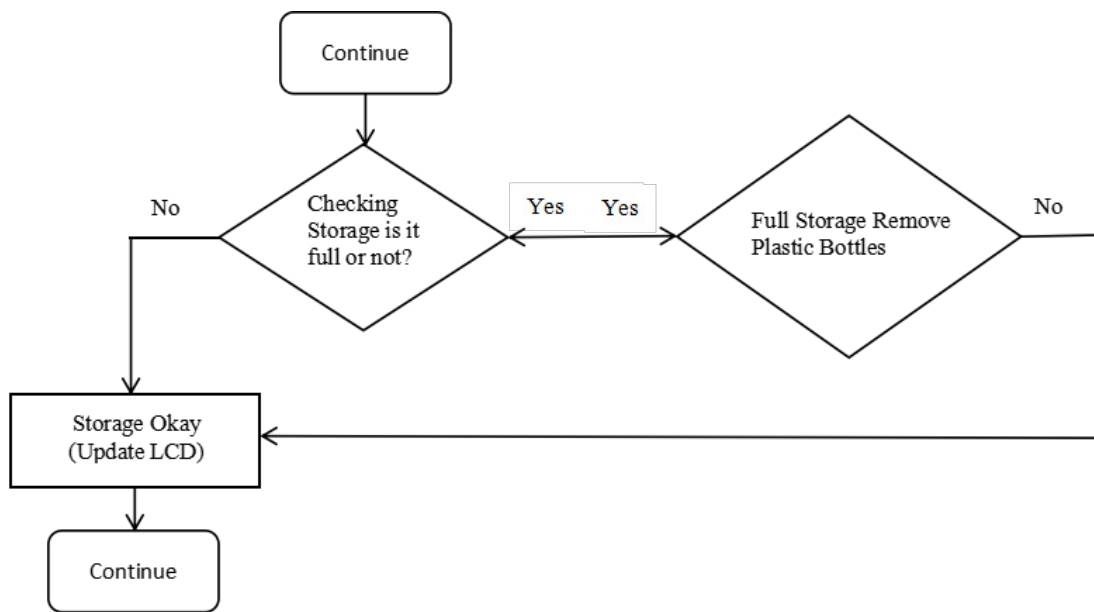


Figure 4.3. Flowchart

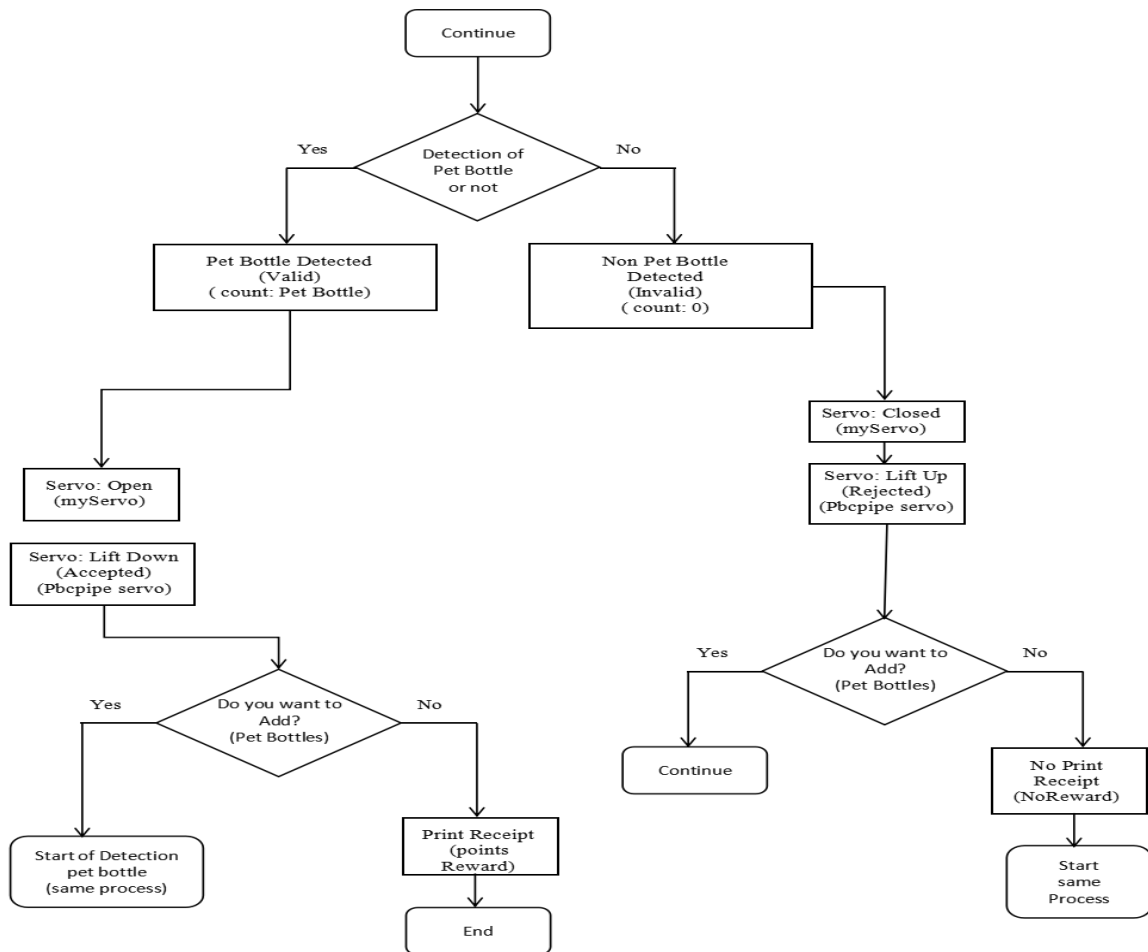


Figure 4.4. Flowchart



Survey Results

The usability test for the Plustic Vendo Machine was conducted on November 3, 2025 at Golden West Colleges. Fifty students from various departments operated the Plustic Vendo Machine in a real-use setting to evaluate ease of use, clarity of instructions, responsiveness, and overall experience. After using the system, participants completed a structured Likert-scale survey. The results provide insight into the system's current performance and identify areas for refinement.

1. The Plustic Vendo Machine was easy to use and understand.

Most students found the machine easy to use, with 50% strongly agreeing and 44% agreeing, showing that users generally understood the process without difficulty. The 6% uncertain group suggests that a few first-time users may need a bit more initial clarity.

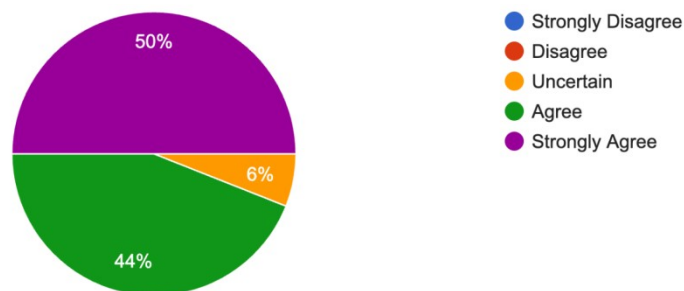


Figure 4.5. Ease of Use

2. The machine's instructions were clear and easy to follow.

Instructions were clear for most respondents, with 54% strongly agreeing and 36% agreeing, indicating that the on-screen guidance worked for the majority. The 10% who were uncertain may represent users who prefer more visual steps or clearer prompts.

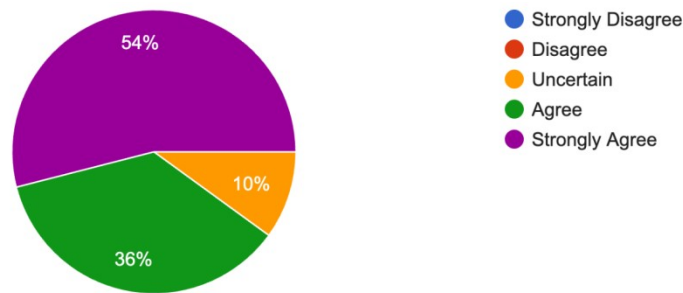


Figure 4.6. Instruction Clarity

3. I was able to operate the machine without needing additional help.

For independent use, 42% strongly agreed and 30% agreed, while 28% were uncertain. This suggests that although most users operated the machine on their own, some may feel more comfortable with brief orientation or on-screen reminders during first use.

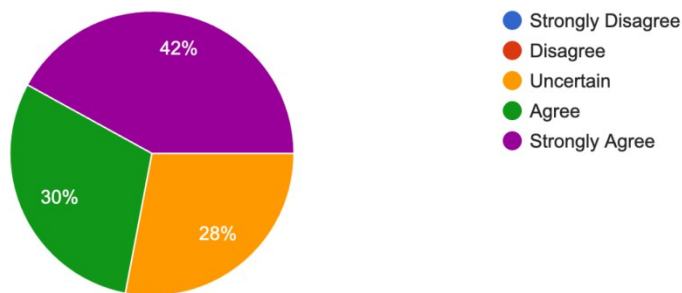


Figure 4.7. User Independence

4. The buttons and display were easy to locate and understand.

Half of the respondents (50% strongly agree and 44% agree) found the buttons and display easy to locate and understand, showing that the interface layout is generally effective. The 6% uncertain response indicates a small group may need more time or familiarity with the controls.

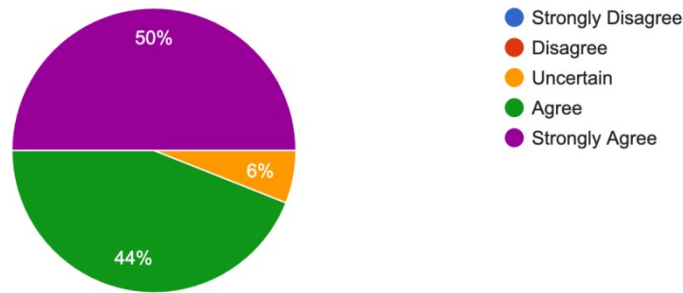


Figure 4.8. Interface Navigation

5. The machine responded quickly to my actions.

Responsiveness received positive feedback, with 42% strongly agreeing and 48% agreeing that the machine reacted in a timely manner. The 10% uncertainty could reflect occasional hesitation or users adjusting to the feedback timing.

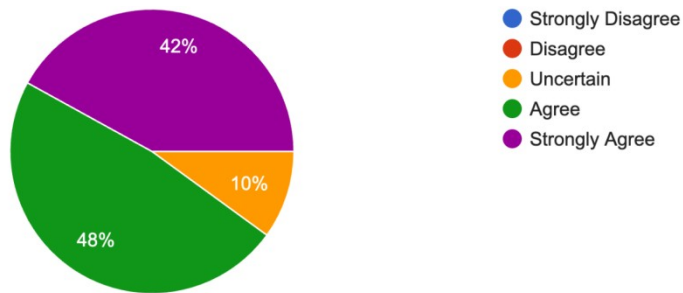


Figure 4.9. System Responsiveness

6. The machine's design is user-friendly and visually appealing.

The machine's design was viewed positively, with 44% strongly agreeing and 48% agreeing that it was user-friendly and visually appealing. The 8% uncertain suggests individual preference differences rather than major design issues.

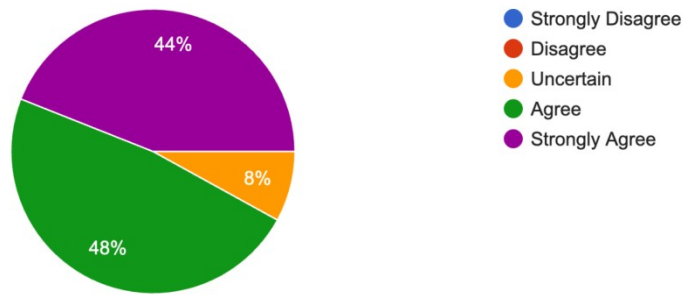


Figure 5.1. Interface Design

7. The screen text and messages are easy to read and understand.

Screen text readability was rated very well, with 46% strongly agreeing and 50% agreeing that on-screen messages were easy to understand. The 4% uncertainty may be related to viewing angles or individual reading preferences.

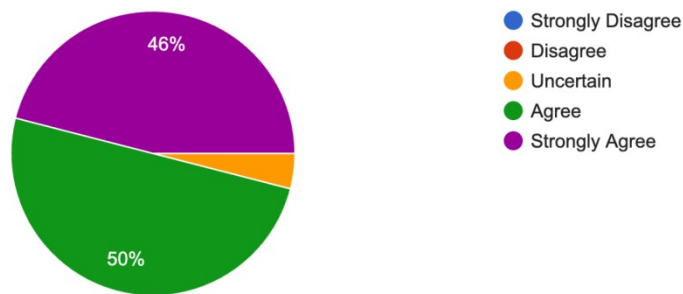


Figure 5.2. Text Readability

8. The machine correctly identified and accepted plastic bottles.

Bottle detection accuracy showed strong results, with 52% strongly agreeing and 40% agreeing, indicating that the recognition function worked as expected for most users. The 8% uncertainty may come from users who did not test multiple bottles or wanted more information on acceptable types.

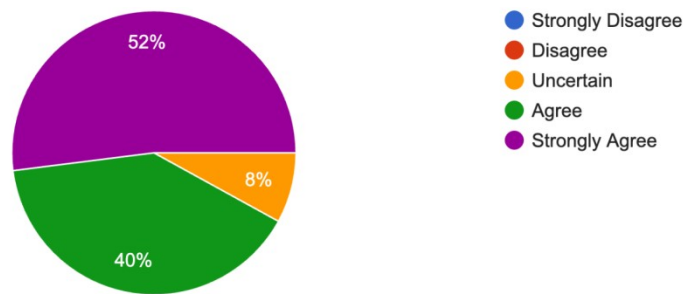


Figure 5.3. Bottle Detection Accuracy

9. The machine maintained consistent performance during my use.

System performance was consistent, with 46% strongly agreeing and 52% agreeing that the machine worked smoothly. Only 2% uncertainty shows that most users experienced steady and reliable operation.

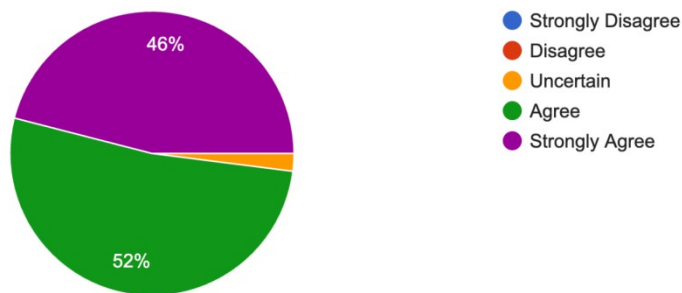


Figure 5.4. Performance Consistency

10. The entire recycling experience was simple and fun.

The overall experience was rated positively, with 58% strongly agreeing and 38% agreeing that the process was simple and enjoyable. The small 4% uncertain suggests that while most users had a straightforward experience, a few may simply need familiarity or repeated use.

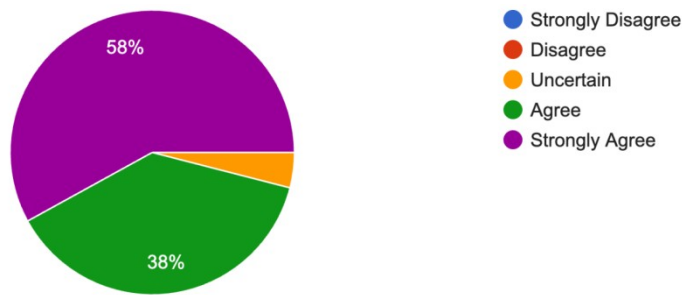


Figure 5.5. Overall User Experience

Device Interface and Features

1. Automatic PET Bottle Detection – The system uses a camera module to identify whether an inserted item is a PET bottle or not, allowing only valid bottles to be accepted for recycling.

2. Servo-Controlled Mechanisms – Several servo motors control different parts of the system, including the main bottle gate, door mechanism, sorting pipe for non-PET items, and a scanning servo that moves the camera.

3. Thermal Printer Output – A built-in thermal printer automatically prints a receipt showing the number of bottles recycled, and time of transaction.

4. Ultrasonic Sensor Storage Check – An ultrasonic sensor measures the distance inside the collection bin to detect if the storage is full, preventing overflow and ensuring proper operation.

5. LCD Display and User Buttons – A 16x2 LCD screen provides clear instructions, real-time status, and feedback messages to guide the user. Four buttons (YES, NO, START, and DOOR) make it easy to interact with the system, start recycling sessions, confirm actions, or close the door manually.



6. Powered by AC and UPS – The machine operates primarily on AC power but includes an Uninterruptible Power Supply (UPS) for backup. This ensures continuous operation or shutdowns during power interruptions.



Figure 5.6. Front View



Figure 5.7. Back View



Figure 5.8. Side View



Figure 5.9. Inside View

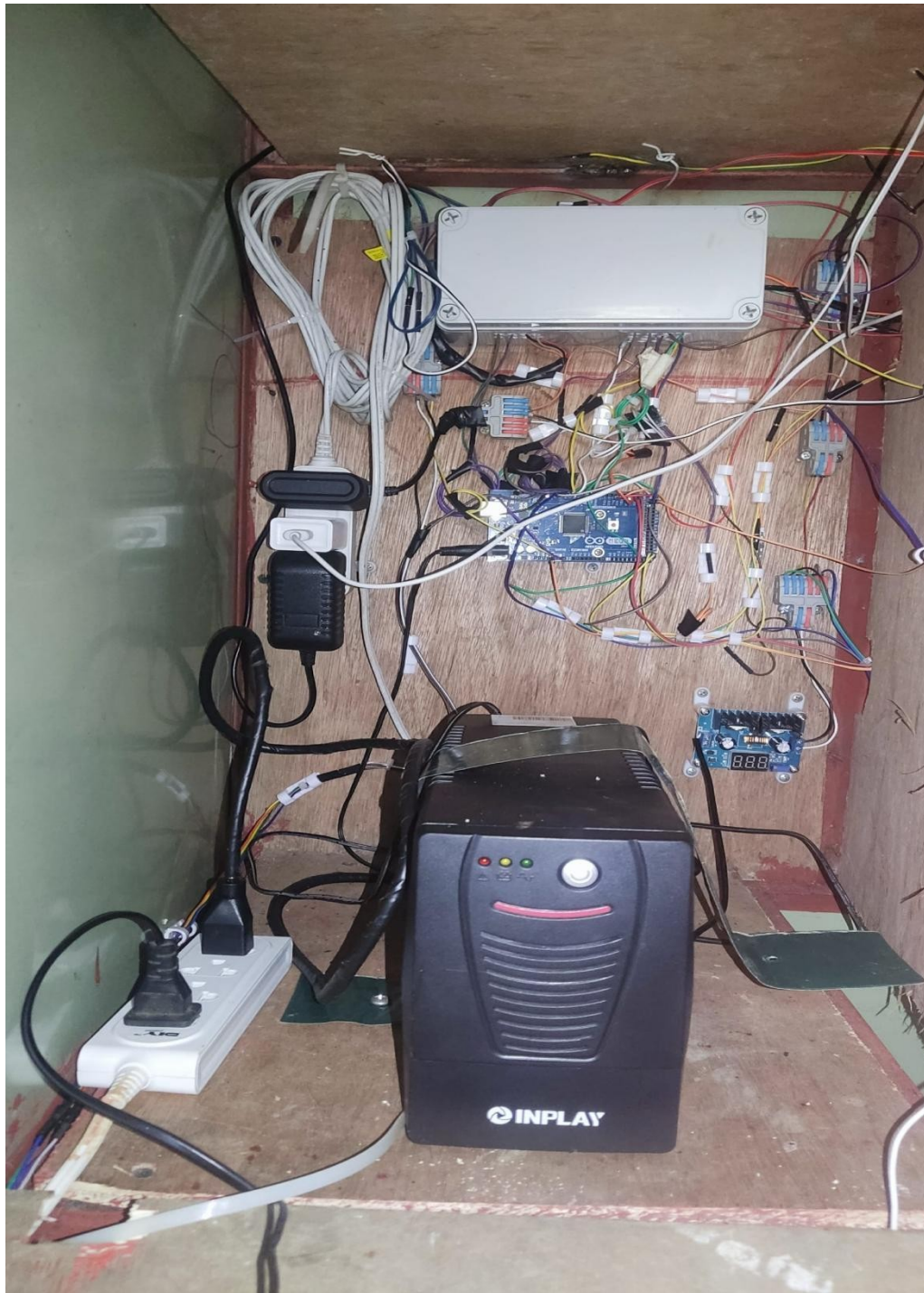


Figure 6.1. Internal Components View

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

The study addressed the problem of rapidly increasing and improper plastic disposal in Golden West Colleges by developing the Plustic Vendo Machine, an Arduino-based Reverse Vending Machine (RVM) designed to automate PET bottle collection and encourage recycling through rewards. The study aimed to identify existing disposal practices, describe the system's features, and assess its usability.

The researchers applied the Rapid Application Development (RAD) model to design and build the prototype. The system consisted of an ESP32-CAM for PET bottle detection, servo motors for gate control, an LCD screen for user guidance, ultrasonic sensors for monitoring storage capacity, and a thermal printer for issuing receipts with reward points.

A usability test was conducted with fifty student participants. Results from the Likert-scale survey showed that most respondents found the machine easy to use, responsive, and user-friendly. These findings indicate that the system functioned as intended and that its reward-based approach helped promote participation in recycling activities. Overall, the project demonstrated a potential method for improving plastic waste collection within the school through the use of automation.

Conclusion

The findings suggest that the Plustic Vendo Machine provided an alternative to the existing manual plastic collection process at Golden West Colleges. The system met its goal of detecting and collecting PET bottles while introducing a reward mechanism that



encouraged proper disposal. Students generally found the system clear and manageable to operate.

The reward feature appeared to motivate some students to recycle more actively, showing that technology-based solutions can support environmental awareness on campus. While the results were positive, further testing in different settings and with more users is recommended to validate the system's performance. Overall, the Plustic Vendo Machine presents a practical approach that may contribute to more organized and efficient recycling practices within the institution.

Recommendations

Reporting. Reporting enables the machine to generate summaries of its performance over a certain period, such as daily, weekly, or monthly. These reports may include the total number of bottles collected, the total rewards dispensed, and system status. It provides valuable insights into usage trends and helps in evaluating the effectiveness of the vending system.

Dispenser. The dispenser feature allows the vending machine to release rewards like coins and tokens after a valid plastic bottle is detected. It acts as a reward mechanism to motivate users to recycle. By automating the dispensing process, the machine creates an engaging and convenient experience for users.

Connectivity. Connectivity introduces the ability for the Plastic Vendo Machine to communicate wirelessly through Wi-Fi or Bluetooth. This feature can be used for remote monitoring, data uploading, or system updates. It enhances flexibility and enables real-time tracking of collected plastics from a distance.

QR Code. The QR code feature allows the system to print or display QR codes representing transaction details, rewards, or user participation. This adds a digital layer to



the recycling process, making it easier to verify records or connect with mobile systems. It also supports transparency and traceability in the recycling program.

Mobile Integration. Adding mobile integration allows the machine to connect with a mobile app or website, giving access to collected data, reward points, and machine status. It helps users and administrators track usage remotely, making the system more modern and interactive. This feature also supports data transparency and long-term sustainability in recycling management.

Emergency Stop Button. The emergency stop button enhances the safety of the vending machine by allowing users or maintenance personnel to immediately stop the system during a malfunction. It prevents hardware damage, ensures safety, and increases system reliability during operation.

Touchscreen Interface. Replacing traditional buttons with a touchscreen makes navigation easier and more modern. Users can select options, view messages, and interact through a visual menu. It enhances the overall design and provides a more user-friendly experience.

Auto Update Feature. If connected to Wi-Fi or mobile data, the system can automatically update its software or firmware. This ensures the machine always runs the latest version with improved features and bug fixes without manual uploading.

Heat Sensor Alert. A temperature sensor can detect overheating inside the machine, especially around motors or electronics. It provides early warnings to prevent fire hazards or component damage.

AI Voice Feedback. AI-generated voice feedback can be used to acknowledge actions (“Plastic detected,” “Thank you for recycling”). It adds personality and makes the machine feel more interactive and human-friendly.



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APPENDIX A

GANTT CHART

Process	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Data Gathering											
Determination Object											
Project Proposal											
Project Development											
Documentation											
Final Defense											
Final Revision											

APPENDIX B



SURVEY QUESTIONNAIRE

Plustic Vendo Machine Questionnaire

Name(Optional):

Date:

Age:

Directions: Please read each statement carefully and select the option that best reflects your experience with the Plastic Vendo Machine.

Use the following scale: **5 – Strongly Agree** **4 – Agree** **3 – Uncertain**
2 – Disagree **1 – Strongly Disagree**

Test Questionnaire	1	2	3	4	5
1. The PlasticVendo Machine was easy to use and understand.					
2. The machine’s instructions were clear and easy to follow.					
3. I was able to operate the machine without needing additional help.					
4. The buttons and display were easy to locate and understand.					
5. The machine responded quickly to my actions.					
6. The machine’s design is user-friendly and visually appealing.					
7. The screen text and messages are easy to read and understand.					
8. The machine correctly identified and accepted plastic bottles.					
9. The system accurately rejected invalid or non-PET bottles.					
10. The LCD displayed accurate and understandable messages.					

APPENDIX C

SAMPLE SURVEY



GOLDEN WEST COLLEGES, INC.

SAN JOSE DRIVE, POBLACION, ALAMINOS CITY, PANGASINAN 2404

Plustic Vendo Machine Questionnaire (Usability Test)

Name(Optional): Juliana Jean Reyes

Date: 11/03/15

Age: 18

Course: BSIT

Directions: Please read each statement carefully and select the option that best reflects your experience with the Plastic Vendo Machine.

Use the following scale: 5 – Strongly Agree 4 – Agree 3 – Uncertain

2 – Disagree 1 – Strongly Disagree

Test Questionnaire	1	2	3	4	5
1. The PlasticVendo Machine was easy to use and understand.					/
2. The machine's instructions were clear and easy to follow.					/
3. I was able to operate the machine without needing additional help.					/
4. The buttons and display were easy to locate and understand.				/	
5. The machine responded quickly to my actions.			/		
6. The machine's design is user-friendly and visually appealing.				/	
7. The screen text and messages are easy to read and understand.					/
8. The machine correctly identified and accepted plastic bottles.					/
9. The machine maintained consistent performance during my use.					
10. The entire recycling experience was simple and fun.				/	



APPENDIX D

SURVEY DOCUMENTATION











APPENDIX E

USER MANUAL

Before using the Plustic Vendo Machine make sure check the storage is not full.

1. Plug the power supply (adapter charger) of the esp32 cam in the extension.
2. Plug the power supply of the lcd, servo's, and the buttons in to the Ups.
3. Plugin the charger of the cctv camera in the extension.
4. Plug the charger of the printer in the extension.
5. Once the charger's Plugin to the extension, Plug the socket of the extension to the Ups.
6. After that plug the socket of the Ups into the 2nd extension and plug to the Ac socket.
7. Power On the Ups to use the Plustic Vendo Machine



APPENDIX F

Source Code

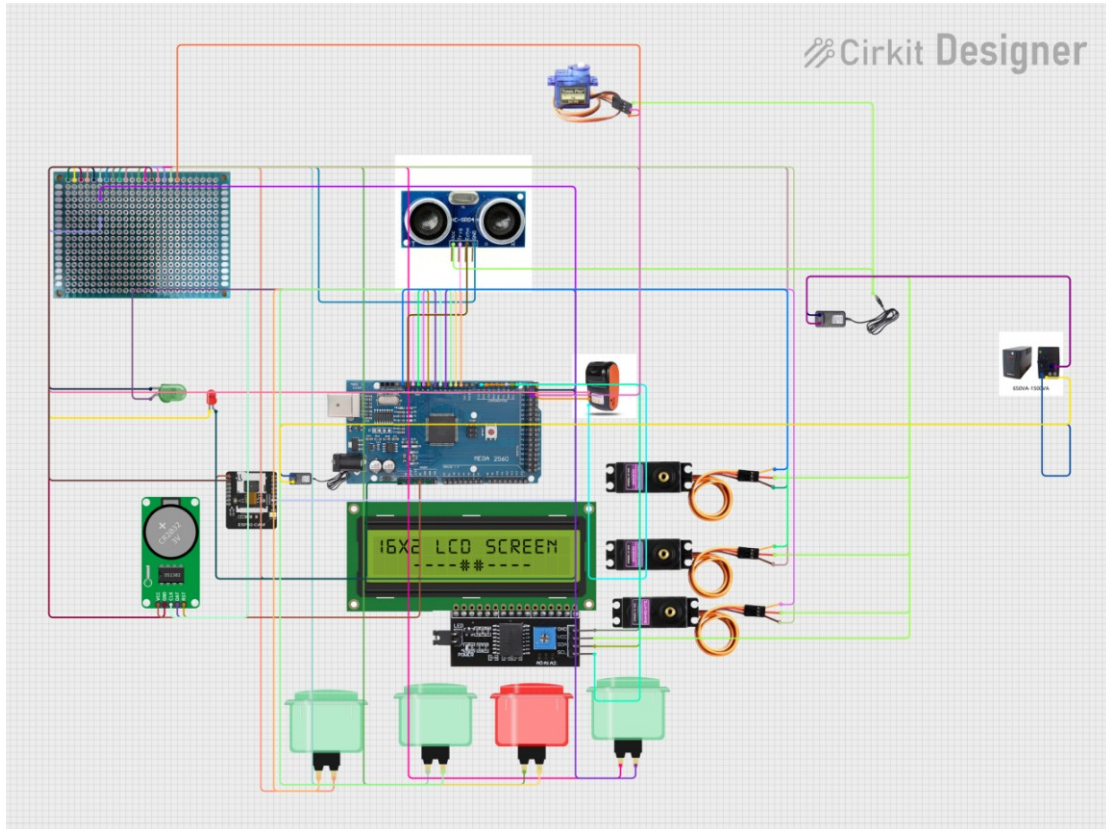


[https://drive.google.com/drive/folders/1XZl5YSXifcXD0x_VR4ywc8HWAZBL89py?
usp=sharing](https://drive.google.com/drive/folders/1XZl5YSXifcXD0x_VR4ywc8HWAZBL89py?usp=sharing)

APPENDIX G



SCHEMATICS





CURRICULUM VITAE

NAME: Mark Darwin R. Garcia

Address: Brgy. Sanjose, Narra St., Alaminos City, Pangasinan

Contact Number: 09608173774

Email Address: darwingarcia686@gmail.com



CAREER OBJECTIVE: To be a part of company that indulges professional growth which provides challenging and rewarding career while allowing me to utilize my knowledge and skills.

EDUCATIONAL BACKGROUND:

Tertiary	:	2-Year Information Technology STI Alaminos 2020- 2022 4 th year Information Technology Gwc 2022 - current
Secondary	:	TVL The Great Plebeian College 2013 - 2019
Primary	:	Pindangan Elementary School 2008-2013

SKILLS AND QUALIFICATIONS:

- Good communication skills
- Basic in using Microsoft Office Tools (MS Word, MS Power point, MS Excel)
- Basic Editing Skills using Photoshop, Canva, Adobe Premiere Pro

RELATED EXPERIENCE

Work Immersion (Heribaco Resort in Alaminos City Pangasinan)
OJT (Lgu Alaminos City Pangasinan)



Date of Birth : November 29, 2001
Place of Birth : Alaminos City Pangasinan
Gender : Male
Civil Status : Single
Nationality : Filipino
Religion : INC
Weight : 53 Kg
Height : 5'5

I hereby certify that the above information is true and correct in the best of my knowledge and belief.

NAME
Applicant

Mark Darwin R. Garcia

NAME: DUCUSIN JAY C.

Address: San Vicente Alaminos City Pangasinan

Contact Number: 09683577306

Email Address: jayducusin00@gmail.com



CAREER OBJECTIVE: To contribute my positive attitude, adaptability, and strong communication skills to support team goals and deliver quality work.

EDUCATIONAL BACKGROUND:

Tertiary	:	4 - Years Information Technology Golden West Colleges Located in Alaminos City, Pangasinan.
Senior High School	:	TVL- HE (Home Economics) The Great Plebeian College 2021-2022
Junior High School	:	San Vicente Elementary School San Vicente Alaminos City Pangasinan 2016-2017

SKILLS AND QUALIFICATIONS:

- Good communication skills
- Knowledgeable in using Microsoft Office Tools (MS Word, MS Power point, MS Excel)

PERSONAL DATA:

Date of Birth	:	December 1, 2003
Place of Birth	:	Quezon City
Gender	:	Male
Civil Status	:	Single
Nationality	:	Filipino
Weight	:	52kg
Height	:	170cm

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

Applicant

Ducusin Jay C.

NAME: JANNA BRIGHT L. ESCAÑO

Address: Mangas, Brgy. Polo, Alaminos City, Pangasinan

Contact Number: 09562918691

Email Address: escanojannabright@gmail.com



CAREER OBJECTIVE: To apply my knowledge in writing, design, and basic technical tools in a practical setting where I can gain experience, contribute to team projects, and develop professional skills.

EDUCATIONAL BACKGROUND:

Tertiary	:	4-Year Computer Science Golden West Colleges San Jose Drive, Alaminos City, Pangasinan 2022 - present
Senior High School	:	Humanities and Social Sciences Alaminos City National High School 2021-2022
Junior High School	:	St. Jerome School of Novaliches T.S. Cruz Subd., Novaliches, QC 2017-2020

SKILLS AND QUALIFICATIONS:

- Can do basic writing tasks, including research papers and documentation.
- Familiar with Microsoft Office tools such as Word, Excel, and PowerPoint.
- Basic knowledge in Canva for creating simple layouts and designs.

PERSONAL DATA:

Date of Birth	:	October 26, 2004
Place of Birth	:	Sta. Cruz, Manila
Gender	:	Female
Civil Status	:	Single
Nationality	:	Filipino
Religion	:	Catholic
Weight	:	45 kg
Height	:	160.02 cm

I hereby certify that the above information is true and correct to the best of my knowledge and belief.

JANNA BRIGHT L. ESCAÑO

Applicant

NAME: LIMUEL S. USON

Address: Camagsingalan Sual, Pang

Contact Number: 09603465297

Email Address: Limueluson21@gmail.com



CAREER OBJECTIVE:

EDUCATIONAL BACKGROUND:

Tertiary	:	Bachelor of Science in Information Technology Golden West Colleges 2022- 2026
Senior High School	:	Tvl-Eim-Electrical Installation & Maintenance 2019-2021
Junior High School	:	Alaminos City National High School 2014-2018

SKILLS AND QUALIFICATIONS:

- Electrical Wiring
- Video Editing (Capcut)
- Sales and marketing skills

PERSONAL DATA:

Date of Birth	:	August 23,2000
Place of Birth	:	Sual Pangasinan
Gender	:	Male
Civil Status	:	Single
Nationality	:	Filipino
Religion	:	Roman Catholic
Weight	:	56
Height	:	5'5

LIMUEL S. USON



GOLDEN WEST COLLEGES, INC.

SAN JOSE DRIVE, POBLACION, ALAMINOS CITY, PANGASINAN 2404

Applicant